

Session 3.

Second order system

Complex poles can be defined with its real and imaginary part

$$s = -\sigma \pm j\omega_d$$

The denominator corresponding to a complex pair will be,

$$\begin{aligned} p(s) &= (s + \sigma - j\omega_d)(s + \sigma + j\omega_d) = (s + \sigma)^2 + \omega_d^2 \quad (i) \\ &= s^2 + 2\sigma s + (\sigma^2 + \omega_d^2) \end{aligned}$$

The transfer function from a second-order differential equation is given by,

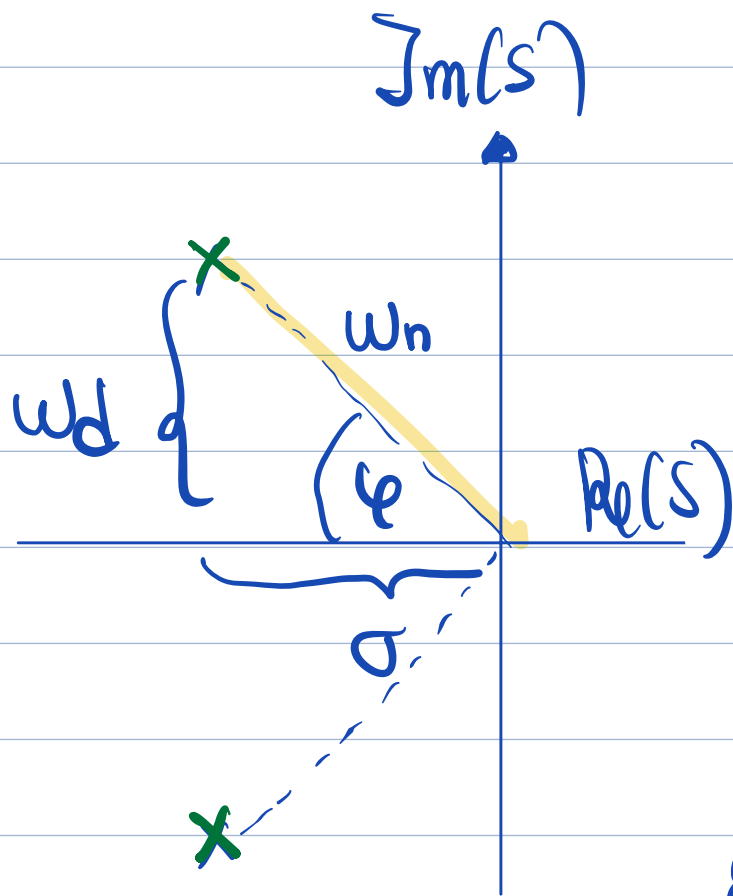
$$H(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \quad (ii)$$

Comparing (i) with (ii)

$$\sigma = \xi \cdot \omega_n \quad (\text{iii})$$

$$\omega_n^2 = \omega_d^2 + \sigma^2$$

$$\omega_n = \sqrt{\omega_d^2 + \sigma^2} \quad (\text{iv})$$



$$\sigma = \xi \cdot \sqrt{\omega_d^2 + \sigma^2} \Rightarrow \xi = \frac{\sigma}{\sqrt{\omega_d^2 + \sigma^2}}$$

$$\cos(\varphi) = \frac{\sigma}{\sqrt{\omega_d^2 + \sigma^2}}$$

$$\cos(\varphi) = \xi$$

where ξ is the damping ratio
and ω_n is the undamped natural freq.

$$\varphi = 90^\circ$$

$$\xi = 0 \Rightarrow \sigma = 0$$

$$\xi = 1 \Rightarrow (s + \omega_n)^2$$

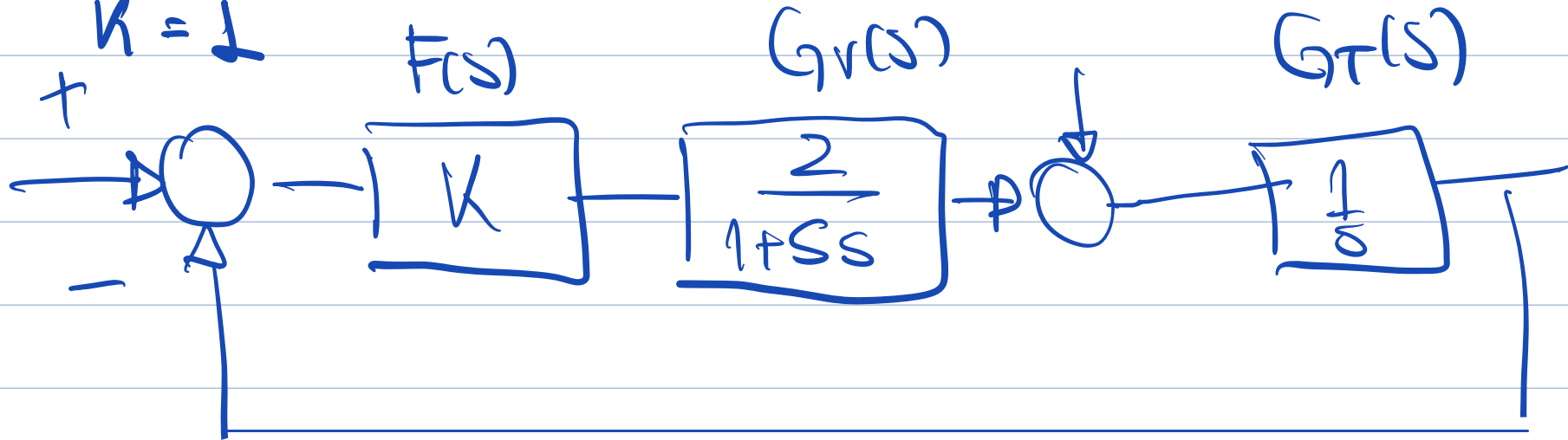
$$s = j\omega_d = j\omega_n$$

$$s = -\omega_n$$

Exercise 3.2.

$$\rightarrow \varphi = 0^\circ$$

a) $K = 1$



Closed loop system poles

$$1 + F(s) \cdot G_v(s) \cdot G_t(s) = 0$$

$$1 + \frac{2}{1+5s} \cdot \frac{1}{s} = 0$$

$$5s^2 + s + 2 = 0$$

$$s = \frac{-1 \pm \sqrt{1 - 40}}{10} = \frac{-1 \pm j\sqrt{39}}{10} = -0.1 \pm j0.62$$

b) Now $F(s) = 1 + K_d \cdot s$ (equation of a pd)

$$1 + F(s) + G_v(s) - G_r(s) = 0$$

$$1 + (1 + K_d s) \cdot \frac{2}{1 + 5s} \cdot \frac{1}{s} = 0 \Rightarrow 5s^2 + s + 2 + 2K_d s = 0$$

$$5s^2 + (1 + 2K_d)s + 2 = 0$$

→ Calculate K_d for $\zeta > 1/\sqrt{2}$

From second order transfer function, we re-write the previous equation

$$s^2 + \frac{1}{5}(1 + 2K_d)s + \frac{2}{5} = 0 \Rightarrow \omega_n = \sqrt{\frac{2}{5}}$$

$$2\zeta \cdot \omega_n = \frac{1}{5}(1 + 2K_d) \Rightarrow \zeta = \sqrt{\frac{5}{2}} \frac{1}{10}(1 + 2K_d) > \frac{1}{\sqrt{2}}$$

$$(1 + 2K_d) > \frac{10}{\sqrt{5}} \Rightarrow K_d > \left(\frac{10}{\sqrt{5}} - 1 \right) \frac{1}{2} = 1.736$$

Root Locus

The root locus is the set of values of s for which,

esto pñero

$$1 + K \cdot G(s) = 0 \Rightarrow 1 + K \cdot \frac{Q(s)}{P(s)} = 0 \quad \text{characteristic}$$

THE BASIC FORM for drawing root locus $P(s)$ equations of the system

In this case the roots on the locus are the closed loop poles of the system for different values of K .

Rules

Rule 1) There are r lines (loci) when n is the degree of $P(s)$ and m is the degree of $Q(s)$. $(n-m)$: lines that go to ∞

Rule 2) As K increase from 0 to ∞ , the roots move from the poles of $G(s)$ to the zeros of $G(s)$

start with it !!

$\times$ = poles of $G(s)$
 $\circ$ = zero of $G(s)$

pole go to the zero!

$$1 + K \frac{Q(s)}{P(s)} = 0 \Rightarrow P(s) + K Q(s) = 0$$

$\therefore s$

the closed loop moves from the open loop

poles of $G(s)$ are when $P(s)=0$
 zeros of $G(s)$ are when $Q(s)=0$

poles of $(1+s)$ to the
 open loop zero of $(1+s)$
 as $K=0 \rightarrow \infty$

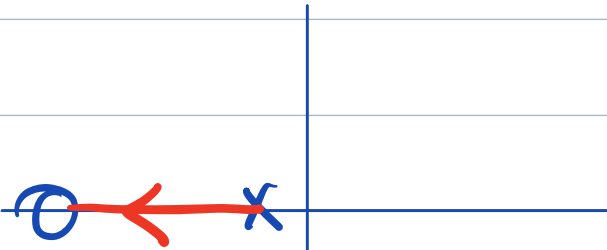
at the beginning of the root locus $K=0$, $P(s)=0 \Rightarrow$
 it starts with the poles of $G(s)$.

$\rightarrow$ As $K \rightarrow \infty$, $P(s) + \infty Q(s)=0$, the
 root locus ends at the zero of $G(s)$, $Q(s)=0$.
 what happens when we have same or different number of
 zeros and poles?

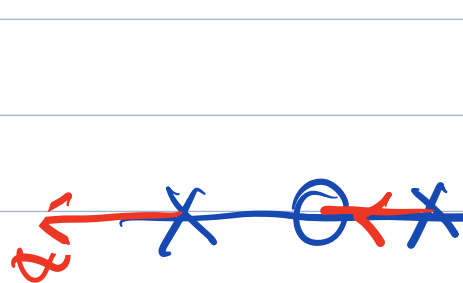
If $\# p_i = \# z_i$

$\# p_i > \# z_i$

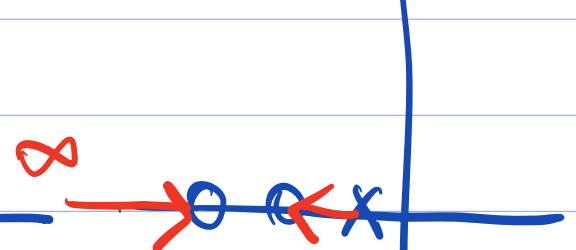
$\# p_i < \# z_i$



from pole to
zero

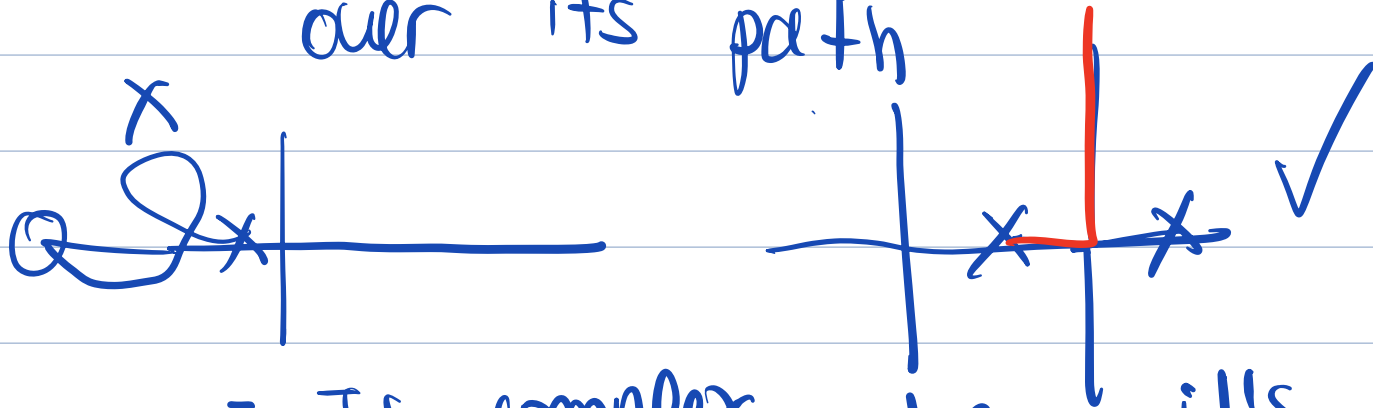


extra poles go
to ∞



extra line comes
from ∞

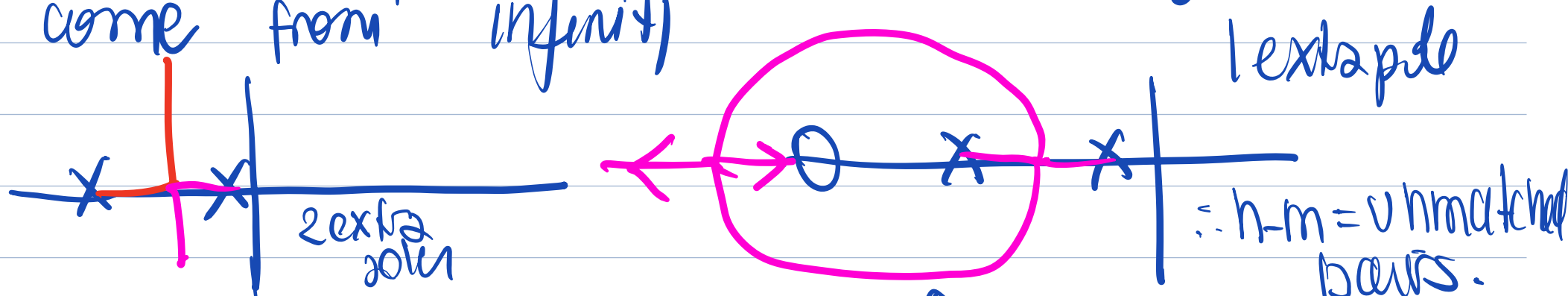
Note: - At no time will be same roots cross over its path



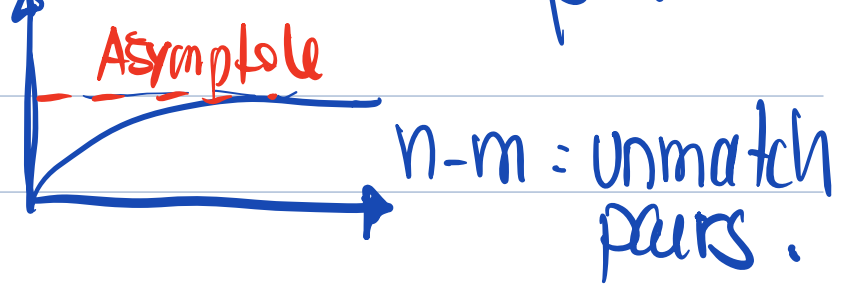
- If complex poles it's symmetric to the real line

Rule 3) Line leave (breakout) and enter (breakin) the real axis at σ_D (we are not calculating this)

Rule 4) If there are not enough poles or zeros to make a pair the extra lines go to or come from infinity



Rule 5) Lines go/come to inf. along asympt. for unmatched zero-pole pairs!
 - Angle of Asymptotes



$$\phi_A = \frac{(2q+1)\pi}{n-m}$$

where $q = 0, 1, \dots, (n-m-1)$
 $\# \text{ Asymp} = n-m$
 $\rightarrow$ The other branches end at a zero

Rule 6)

- The centroid of the asymptotes:
 all come together to a single point on the real axis

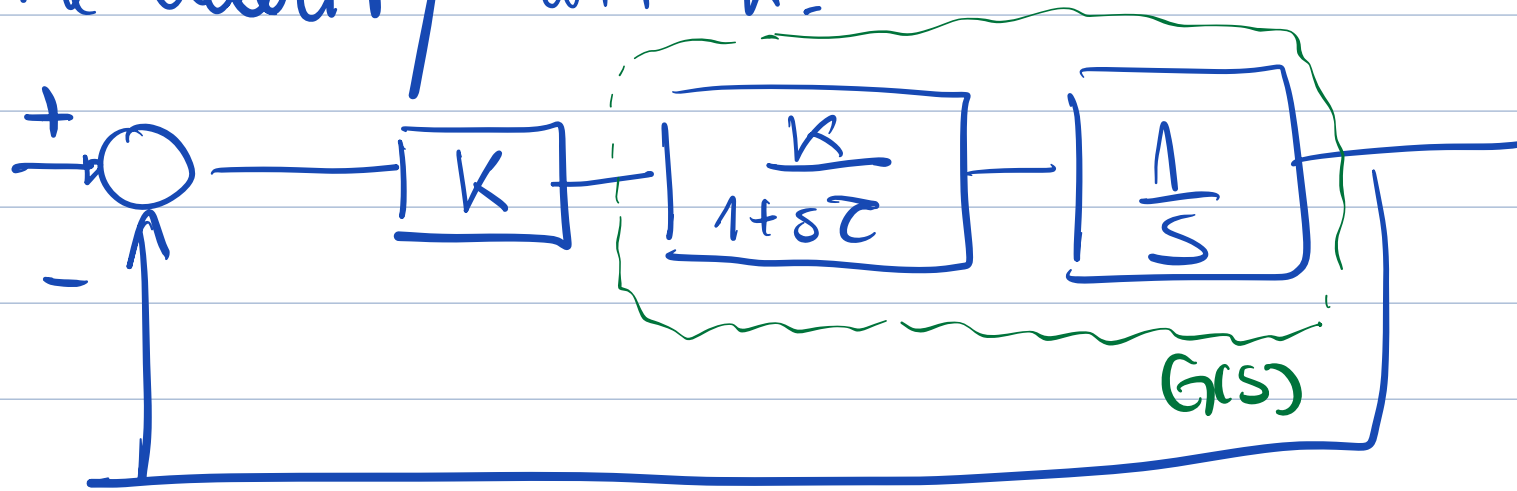
$$\frac{\sum p_i - \sum z_i}{n-m}$$

DRAW SKETCH :)

Rule 7) Intersection with the imaginary axis:
 Insert $s = i\omega$ into $1 + K \cdot G(s) = 0$

make $\omega = 0$ (we want to be at the axis) $\Rightarrow$ For that K the locus pass by Im axis

3.6 a) Draw root locus when we can't measure the velocity wrt K .



1) Let's write in the form : $1 + K \cdot G(s) = 0$ (1)

$$1 + K \cdot \frac{2}{(1 + \frac{1}{2}s)} \cdot \frac{1}{s} = 0 \Rightarrow 1 + K \frac{4}{(s^2 + 2s)} = 0$$

$$G(s) = \frac{4}{s(s+2)} = \frac{Q(s)}{P(s)} \Rightarrow \begin{array}{l} m=0 \\ n=2 \end{array} \quad \begin{array}{l} + n > m \text{ * (2 lines (loci))} \\ + 2 \text{ unmatched poles} \end{array}$$

2) Let's write (1) as $s^2 + 2s + 4K = 0$
 For $K=0$ $s^2 + 2s = 0 \rightarrow s=0 \quad s=-2$

For $K=\infty$. No zeros on G(s), then we know poles has to go to infinity or 2 asymptotes

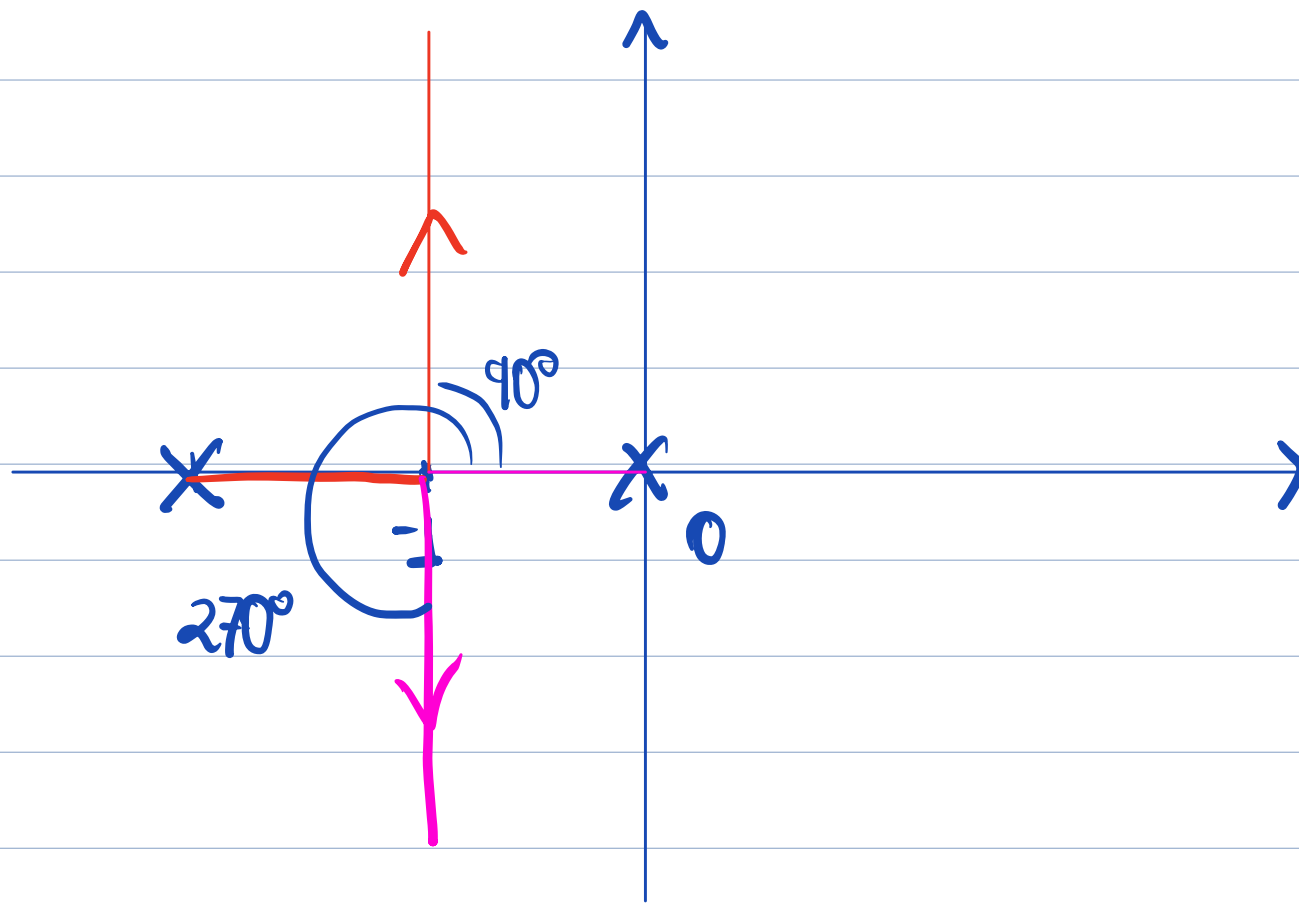
ANGLES OF ASYMPTOTE

3) $\phi_A = \frac{(2q+1)\pi}{n-m}$ for $q = 0, 1, \dots, (n-m-1) = 0, 1$

$\phi_{A1} = \frac{1}{2}\pi \quad \phi_{A2} = \frac{3}{2}\pi$

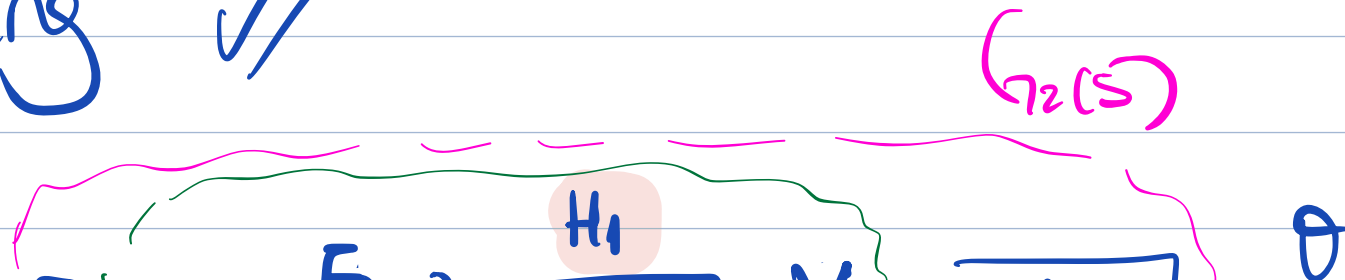
4) centroid of asymptotes (where they go to ϕ_{A1} & ϕ_{A2} ?)

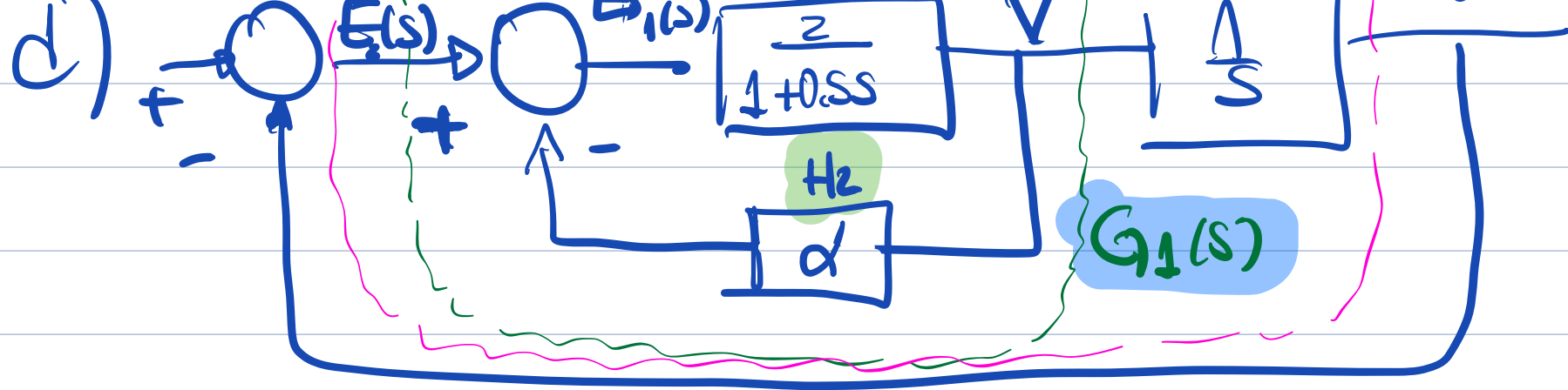
$\frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{(0 + -2)}{2} = -1 //$



s) Intersection - $s^2 + 2s + 4K = 0 \quad | s = j\omega$

$$-\omega^2 + 2j\omega + 4K = 0 \Rightarrow \omega = 0 \Rightarrow K = 0, \text{ only crossing} \quad \checkmark$$





$$G_1(s) = \frac{V(s)}{E_2(s)} \Rightarrow E_2(s) - V(s) \cdot \alpha(s) = V(s) \cdot \left(\frac{1+0.5s}{2} \right)$$

$$E_2(s) = V(s) \left(\alpha + \frac{1+0.5s}{2} \right)$$

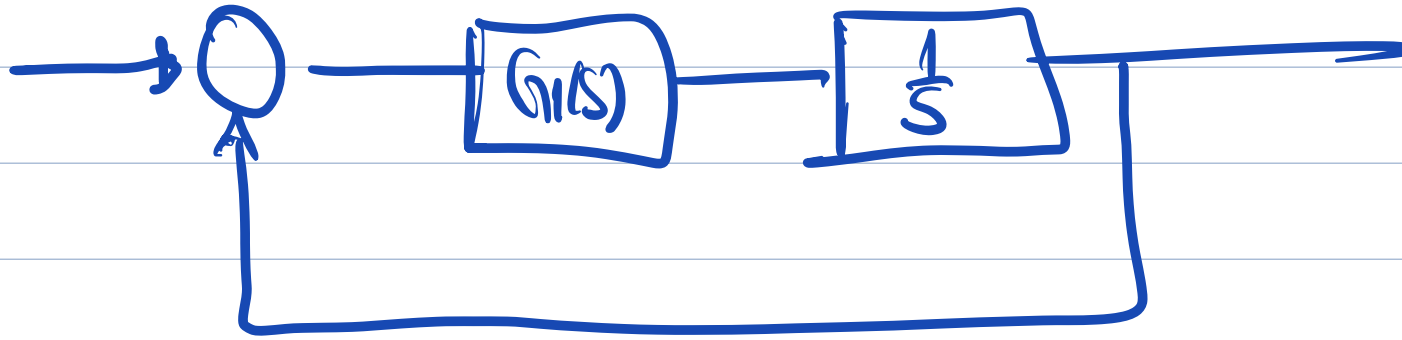
$$\Rightarrow \frac{V(s)}{E(s)} = \frac{2}{(0.5s + 2\alpha + 1)}$$

Recap

$$\Rightarrow G(s) = \frac{G_{OL}}{1+G_{OL}} \xrightarrow{??} \text{For inner feedback} = \frac{G_{OL}}{\text{poles closed loop system}} = \frac{H_1}{1+H_1 \cdot H_2}$$

does not work!

$$G_1(s) = \frac{2}{1+0.5s} \cdot \frac{1}{1+\alpha \cdot \frac{2}{1+0.5s}} = \frac{2}{0.5s + 2\alpha + 1}$$



$$G_2(s) = \frac{G_1(s) \cdot \frac{1}{s}}{1 + G_1(s) \cdot \frac{1}{s}} = \frac{2}{0.5s + 2\alpha + 1} \cdot \frac{1}{1 + \frac{1}{s} \cdot \frac{2}{0.5s + 2\alpha + 1}}$$

closed loop transfer

$$G_2(s) = \frac{2}{1 + G_1(s) \cdot \frac{1}{s}}$$

$$0.5s^2 + (2\alpha + 1)s + 2$$

$$1 + K \cdot G(s) = 0$$

↳ two complex poles.

$$1 + G_1(s) \frac{1}{s} = 1 + \frac{1}{s} \frac{2}{(0.5s + 2\alpha + 1)} = 0.5s^2 + (2\alpha + 1)s + 2$$

Root Locus

1) Let's write it the Locus form $(1 + KG) =$

$$1 + K \cdot G(s) = 0 \Rightarrow s^2 + 4\alpha s + 2s + 4 = 0$$

$$1 + \alpha \cdot \frac{4s}{(s^2 + 2s + 4)} = 0 \quad \frac{s^2 + 2s + 4 + 4\alpha s}{(s^2 + 2s + 4)} = 0 \quad \frac{1}{(s^2 + 2s + 4)}$$

$$\left. \begin{array}{l} Q(s) = 4(s) \\ P(s) = s^2 + 2s + 4 \end{array} \right\} \begin{array}{l} n = 2 \\ m = 1 \end{array}$$

Then,
unmatched poles
 $n - m = 1$ (1 asymptote)
 $\max(2, 1) = 2$ loci

$$2) \quad s=0 \Rightarrow P(s)=0 \Rightarrow s^2+2s+4=0$$

$$s = \frac{-2 \pm \sqrt{4-16}}{2}$$

$$s = -1 \pm \sqrt{3}j$$

$$s=0 \Rightarrow Q(s)=0$$

$$s=0 //$$

The other goes to ∞ .

$$3) \quad \phi_A = \frac{(2q+1)\pi}{m-n} \quad q=0 \Rightarrow \phi_A = \pi$$

$$4) \quad \frac{\sum_{\text{poles}} - \sum_{\text{zeros}}}{1} = -1 + \sqrt{3}j + -1 - \sqrt{3}j - 0$$

$$= -2 //$$

5) Intersection with Im axis

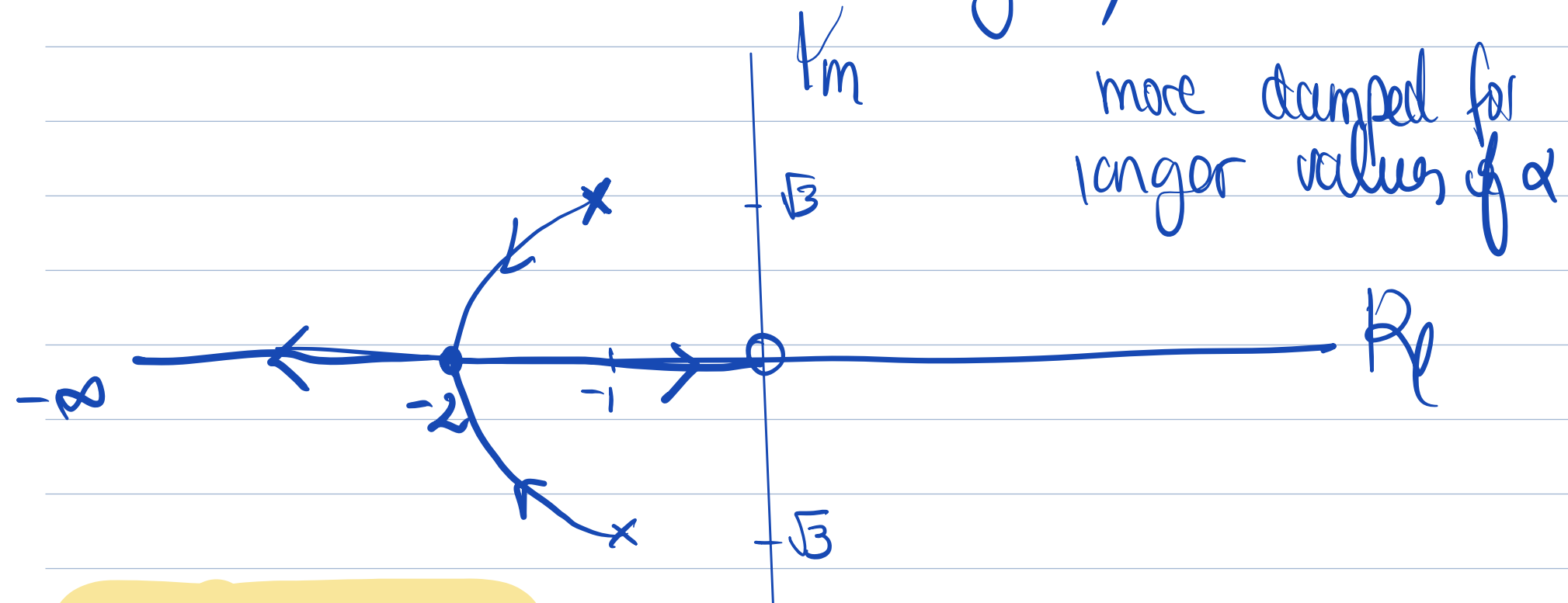
$$1 + K G(s) \Big|_{s=j\omega} = -\omega^2 + 4\alpha j\omega + 2j\omega + 4 = 0$$

$$-\omega^2 + 4 = 0 \quad \& \quad 4\alpha\omega + 2\omega = 0$$

$$\omega = \pm 2$$

$$\alpha = -\frac{1}{2} \quad \times$$

there no α positive for this
 $\Rightarrow$ No intersection with imaginary axis



2021 EXAM

2.b

$$G_1 = \frac{1}{s(s^2 + 6s + 45)} =$$

$$1 + K \cdot \frac{1}{s(s^2 + 6s + 45)} = 0 \Rightarrow \text{For Root Locus sketch.}$$

$$\text{For } K=0 \quad s(s^2 + 6s + 45) = 0$$

$$s_1 = 0 \quad s = \frac{-6 \pm \sqrt{36 - 180}}{2} \Rightarrow 144$$

$$s_{2,3} = -3 \pm j6$$

For $K = \infty$

For $K = \infty$. No zeros on G(s), then we know poles has to go to infinity.

Asymptotes: $n - m = 3 - 0 = 3$

Angles: $\phi_A = \frac{(2q+1)\pi}{n-m}$ for $q = 0, 1, \dots, n-m-1$

$$q_0 = \frac{\pi}{3} = 60^\circ \quad q_1 = \frac{\pi}{180^\circ} \quad q_2 = \frac{5\pi}{3} = 300^\circ = -60^\circ$$

$$\begin{aligned} \text{Centroid: } \frac{\sum \text{poles} - \sum \text{zeros}}{n-m} &= \frac{-3 + \cancel{6j} - 3 - \cancel{6j} + 0}{3} \\ &= -2 \end{aligned}$$

Cross with imaginary Axis

$$1 + K \frac{1}{s(s^2 + 6s + 45)} = 0$$

$$s^3 + 6s^2 + 45s + K = 0 \quad / \quad s = j\omega$$

$$-j\omega^3 - 6\omega^2 + 45j\omega + K = 0$$

$$\text{Real part} = 0 \Rightarrow K - 6\omega^2 = 0$$

Im. part = 0 $\Rightarrow$

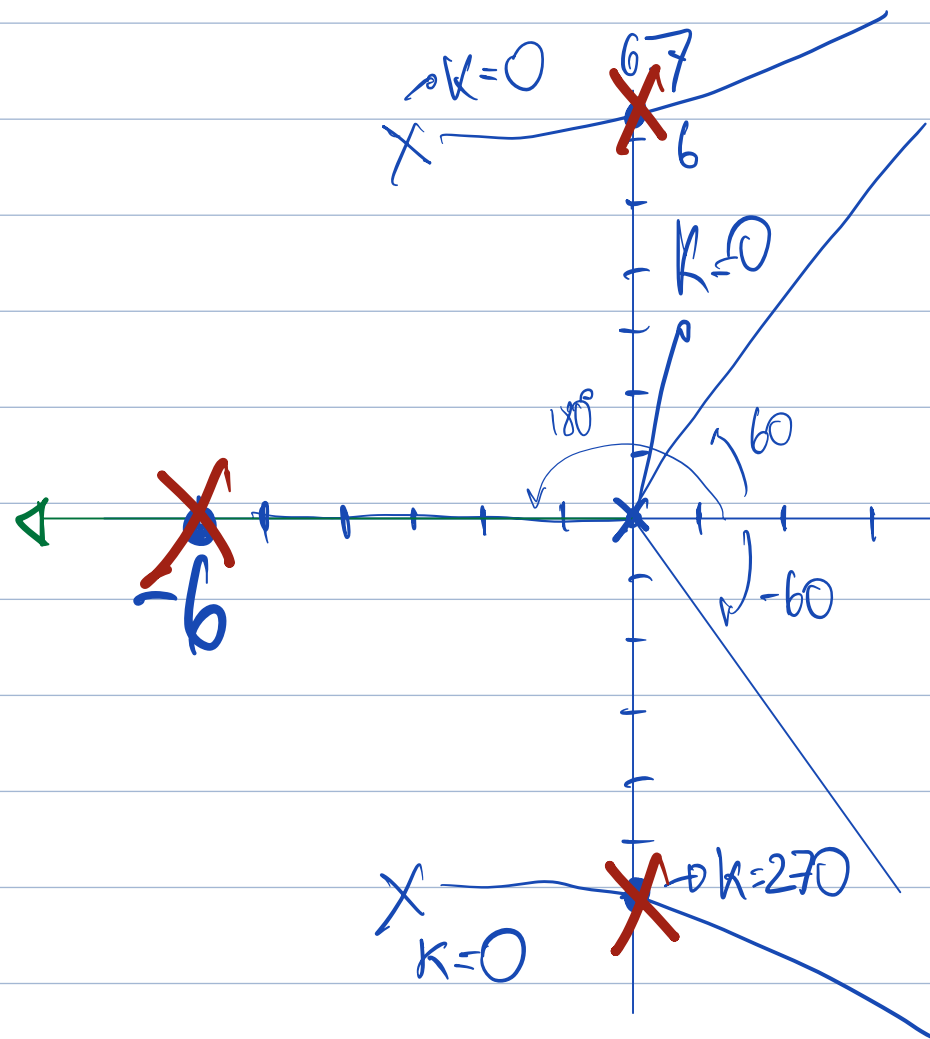
$$-j\omega^3 + 45j\omega = 0 \quad \} \quad \omega(-\omega^2 + 45)j = 0$$

$$\omega = 0 \quad \omega = \sqrt{45}$$

$$K = 0$$

$$K = 270$$

Do you want to know exactly which are the poles for $K = 270$? solve

$$s^3 + 6s^2 + 45s + 270 = 0$$


$$s = -6 \quad s = \pm 3i\sqrt{5} = 6.7j$$

∴ For stable system
 $0 < K < 270$

System 2

$$G_2 = \frac{0.075s^2 + s + 1}{s(s^2 + 3s + 5)} \Rightarrow 1 + K \frac{P(s)}{Q(s)} = 0$$

Asympt. = $3 - 2 = 1$!

$\phi_A = \pi$

$K=0 \Rightarrow P(s)=0 \Rightarrow s(s^2 + 3s + 5) = 0$

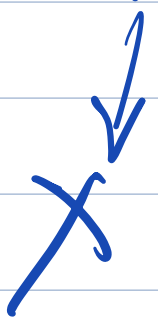
$s=0 \quad s = \frac{-3 \pm \sqrt{9 - 20}}{2} = -1.5 \pm j^{1.66}$

$K=\infty \quad Q(s)=0$

$s = \frac{-1 \pm \sqrt{1 - 0.3}}{0.15} = \begin{cases} -1.08 \\ -12.24 \end{cases}$

Centroid

$$\frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{-3 + 13.32}{1} = 10.32$$



Axis Im. cross

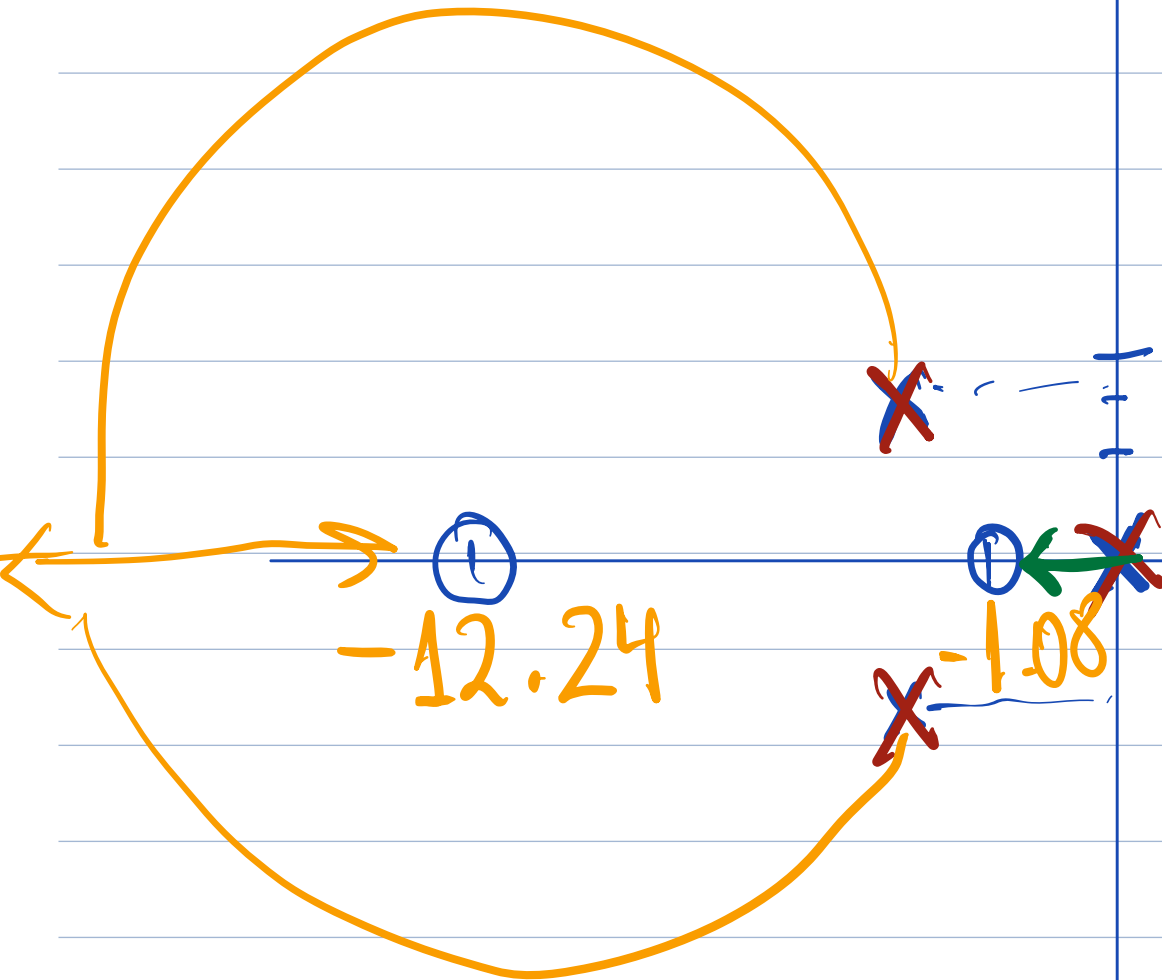
$$s^3 + 3s^2 + 5s + K \cdot 0.07s^2 + K \cdot s + K = 0$$

$$-j\omega^3 - 3\omega^2 + 5j\omega - K \cdot 0.07s\omega^2 + Kj\omega + K = 0$$

$$-3\omega^2 - K \cdot 0.07s\omega^2 + K = 0$$

$$-\omega^3 + 5\omega + K\omega = 0 \Rightarrow \omega \neq 0 \Rightarrow K = 0$$

$$K > 0$$



$$K = 40$$

$$H_{cl} = \frac{G \cdot K}{1 + GK}$$

$$H_{cl_1} = \frac{\frac{40}{s(s^2 + 6s + 45)}}{1 + \frac{40}{s(s^2 + 6s + 45)}} = \frac{40}{s^3 + 6s^2 + 45s + 40}$$

$\hookrightarrow (s+1)(s^2 + 5s + 40)$

$$H_{cl_2} = \frac{40 \frac{0.075s^2 + s + 1}{s(s^2 + 3s + 5)}}{1 + \frac{(0.075s^2 + s + 1) \cdot 40}{s(s^2 + 3s + 5)}} = \frac{6s^2 + 40s + 40}{s^3 + 6s^2 + 45s + 40}$$

$H_{cl1} : s = -1$ → slow pole
 $s = \frac{-5 \pm \sqrt{25 - 160}}{2} \Rightarrow -2.5 \pm j\omega$

∴ Dominant is $s = -1$

$$H_{cl2} = \frac{40 \cdot (0.075s^2 + s + 1)}{(s+1)(s^2 + 5s + 40)} = \frac{40 \cdot [0.075(s+4.1)(s+12.2)]}{(s+1)(s^2 + 5s + 40)}$$

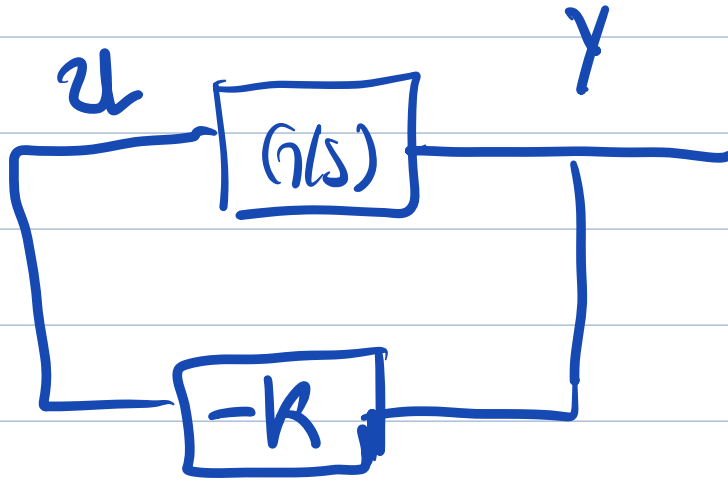
$$\approx \frac{6(s+12.2)}{(s^2 + 5s + 40)}$$

~~zero~~ = -12.2
 poles = $\frac{-5 \pm \sqrt{25 - 160}}{2} = -2.5 \pm j\omega$

G_1 with 1 - dominant pole shape
the system behaves

G_2 with 2 - oscillations overshoot
and damping in ss.

3.22



$$U \cdot G(s) = Y$$

$$U = -KY$$

$$-KY \cdot G(s) = Y$$

$$(K \cdot G(s) + 1) \cdot Y = 0$$

For $Y(s) \neq 0$

$$1 + K \cdot G(s) = 0 \quad (\text{closed loop poles})$$

$$G(s) = \frac{P_{n-1}(s)}{Q_n(s)} \Rightarrow Q_n(s) + K T_{n-1}(s) = 0$$

All zeros are in the LHP.

$$\text{If } K=0$$

$Q_n(s) \Rightarrow \text{poles of } G(s)$

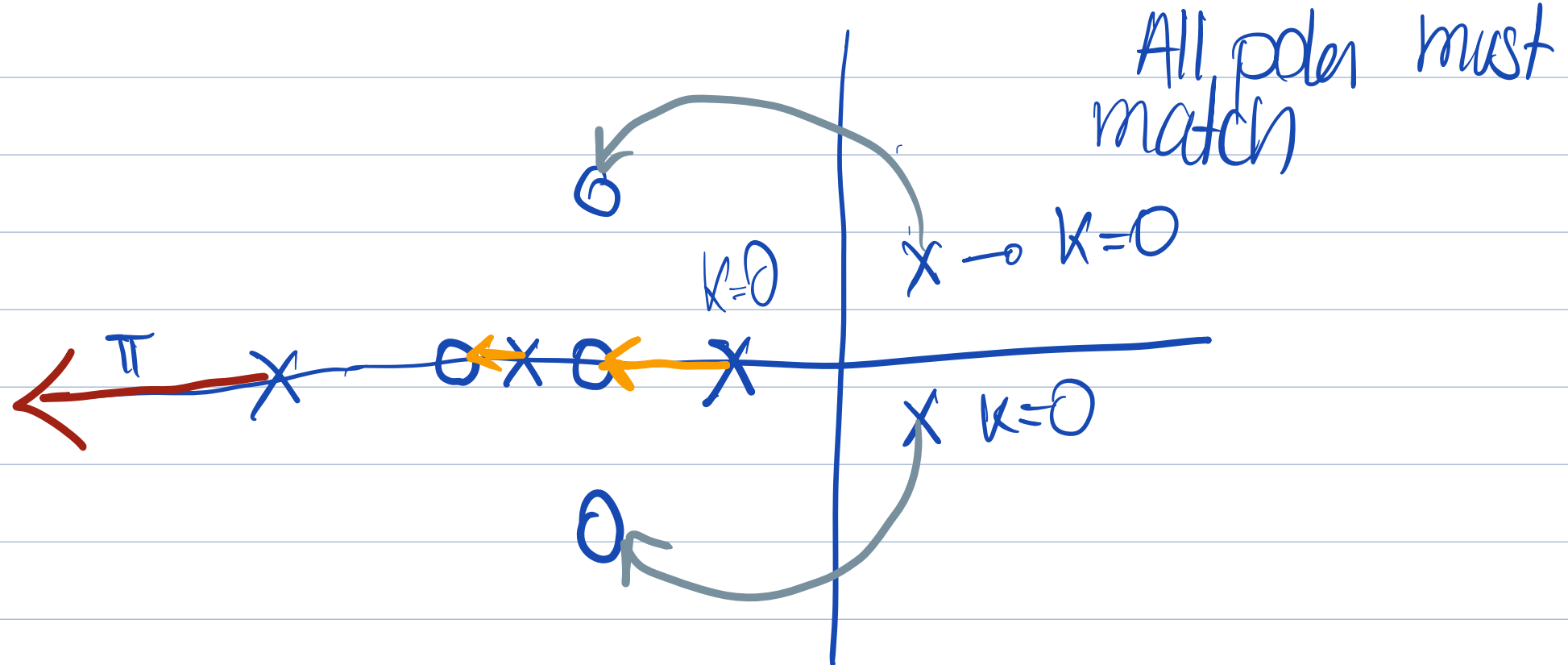
$$\text{If } K=\infty$$

$T_{n-1}(s) \Rightarrow \text{zeros of } G(s)$

But $n = n$ $m = n-1$, then

$$m-n = -1 \Rightarrow \phi_A = \frac{(2q+1)\pi}{m-n} = \pi$$

When $K \rightarrow \infty$, one pole goes to infinity and the remaining poles match the zero on the LHP! Therefore if K is big enough the system will have all the poles in the LHP.



$k=0$ unstable, but $k \rightarrow \infty$ stable